

Let $\vec{\Gamma} = (\Gamma_1, \dots, \Gamma_n)$ be a random vector. Consider:

- $\vec{m} := (\mathbb{E}[\Gamma_1], \dots, \mathbb{E}[\Gamma_n])$.
- $\Sigma \in M(n \times n, \mathbb{R})$ such that $\Sigma(i, j) = \text{Cov}(\Gamma_i, \Gamma_j) = \mathbb{E}[(\Gamma_i - m_i)(\Gamma_j - m_j)]$.

$\vec{\Gamma}$ is Gaussian iff we can write $\phi_{\vec{\Gamma}}(\vec{\xi}) = \exp(i\mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}] - \frac{\text{Var}(\vec{\xi} \cdot \vec{\Gamma})}{2})$.

Teorema 1. • $\mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}] = \vec{\xi} \cdot \vec{m}$

- $\text{Var}(\vec{\xi} \cdot \vec{\Gamma}) = \vec{\xi}^T \cdot \Sigma \cdot \vec{\xi}$

Proof.

$$\begin{aligned} \mathbb{E}[\vec{\xi} \cdot \vec{\Gamma}] &= \mathbb{E}\left[\sum_{j=1, \dots, n} \xi_j \cdot \Gamma_j\right] \\ &= \sum_{j=1, \dots, n} \xi_j \cdot \mathbb{E}[\Gamma_j] = \sum_{j=1, \dots, n} \xi_j \cdot m_j = \vec{\xi} \cdot \vec{m} \end{aligned}$$

$$\begin{aligned} \text{Var}(\vec{\xi} \cdot \vec{\Gamma}) &= \mathbb{E}[(\vec{\xi} \cdot \vec{\Gamma})^2] - (\vec{\xi} \cdot \vec{m})^2 \\ &= \vec{\xi}^T \cdot \Sigma \cdot \vec{\xi} \end{aligned}$$

where the last equality follows because of the following:

$$\begin{aligned} \mathbb{E}[(\vec{\xi} \cdot \vec{\Gamma})^2] &= \mathbb{E}\left[\sum_{j=1, \dots, n} \sum_{k=1, \dots, n} \xi_j \Gamma_j \xi_k \Gamma_k\right] \\ &= \mathbb{E}[\vec{\xi}^T \cdot \Gamma \cdot \Gamma^T \cdot \xi] \\ &= \vec{\xi}^T \cdot \mathbb{E}[\vec{\Gamma} \cdot \vec{\Gamma}^T] \cdot \xi \end{aligned}$$

$$\mathbb{E}[\vec{\Gamma} \cdot \vec{\Gamma}^T](i, j) = \mathbb{E}[\Gamma_i \cdot \Gamma_j] = \Sigma(i, j) + m_i m_j$$

□